

Alex Klee

alex.klee111@gmail.com | linkedin.com/in/alex-klee | abklee.com

EDUCATION

Virginia Tech, College of Engineering | College of Science **Blacksburg, VA**
BS Computer Science | BS Applied Discrete Mathematics | Minor: Cybersecurity Aug 2021 - May 2025

GPA: 3.9/4.0

Virginia Tech, College of Engineering **Blacksburg, VA**
Accelerated Master of Engineering, Computer Science and Applications Aug 2024 - May 2026

PROFESSIONAL EXPERIENCE

Peraton **Blacksburg, VA**
Data Science Engineer Intern Jun 2025 - Present

- Researched and applied techniques for hardening a custom MCP server for secure AI agent execution across domains
- Designed agentic AI workflows for different tasks, e.g. scanning cloud-native codebases for compliance / controls
- Integrated AI-driven telemetry analysis with monitoring platforms to automate anomaly detection across K8s clusters
- Collaborated within a SAFe Agile team structure, coordinating across teams to deliver production-grade features

Hume Center for National Security and Technology at Virginia Tech **Blacksburg, VA**
HIRF Research Fellow August 2024 - May 2025

- Research project with industry partner L3Harris focused on software reverse engineering and vulnerability detection
- Investigated potential vulnerabilities (primarily remote code execution and privilege escalation) of various messaging apps on iPhones through reverse engineering, patching, sideloading, and dynamic analysis
- Leveraged tools such as Ghidra, x64dbg, and many iOS-related tools to effectively analyze iOS binaries

Hume Center for National Security and Technology at Virginia Tech **Arlington, VA**
Intern May 2024 - July 2024

- Explored, tested, and evaluated various applications of quantum optimization and quantum random numbers across different fields including machine learning, cybersecurity, and cryptography
- Found advantages of quantum optimization technology in multiple machine learning algorithms using PyTorch
- Tested and analyzed effectiveness of quantum random numbers in different cryptosystems

Hume Center for National Security and Technology at Virginia Tech **Blacksburg, VA**
Student Researcher (Software Team) Jan 2024 - May 2024

- Researcher on project Defend the Republic with the goal of developing a lighter-than-air automated drone
- Automated and tested our team's drone using OpenCV in C++ and Python for computer vision and ROS2

Virginia Tech Department of Computer Science **Blacksburg, VA**
Undergraduate Teaching Assistant Aug 2023 - Dec 2023

- UTA for CS1114 Intro to Software Design, a class focused on the foundations of object oriented programming in Java
- Held labs and office hours for students in the class multiple times a week, teaching foundations of Java and OOP

PROJECTS

Assessing the Vulnerability of LoRA-Based Fine-Tuning to Data Poisoning Attacks **Jan 2025 – May 2025**
Senior CS Capstone Project

- Researched security risks of LoRA-based fine-tuning of LLMs, specifically Meta LLaMA, across 4 PEFT methods
- Evaluated the impact of data poisoning attacks on 4 different LoRA-based Parameter-Efficient Fine-Tuning methods
- Fine tuned each model with the Beavertails dataset, analyzed results using various metrics such as the ROUGE-F1
- Investigated mitigation strategies to enhance the robustness of fine-tuned models against adversarial manipulation

SKILLS

Languages, Frameworks, etc: Python, Java, C, C++, Linux/Unix, PyTorch, TensorFlow, SQL, LangChain, Docker, Kubernetes, Helm, Terraform, AWS, Ghidra, Bash, HTML, CSS, JavaScript, MongoDB, React, Express.js, Node.js